PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Design features that reduce the use of seclusion and restraint in mental health facilities: A rapid systematic review.
AUTHORS	Oostermeijer, Sanne; Minshall, Catherine; Harvey, Carol; Hamilton, Bridget; Roper, Cath; Martel, Andrew; Fletcher, Justine; Brophy, Lisa

VERSION 1 – REVIEW

REVIEWER	Langan, Julie University of Glasgow, Institute of Health and Wellbeing
REVIEW RETURNED	12-Dec-2020

GENERAL COMMENTS	This was a very interesting paper about ward design and how it may impact on rates of seclusion and restraint. It is clinical relevant. They are a number of strengths of paper but there are also a number of limitations that should be addressed prior to publication. The methods were not completely clear. What are the inclusion and exclusion criteria for the studies, other than non English language? Would it be helpful to include a summary table of the 35 included studies, such as country, ward type (adult, LD etc), study design etc It looks like (from the results) that adult wards and CAMHS wards were included. But were PICU's, Older Adults Wards, Forensic and LD wards included too? I think more description about the 35 studies would be helpful as they are clearly very varied. I wonder if there was any comment on ward occupancy as this is also relevant. It would was also unclear how the the over-arching themes were identified? Is this a rapid systematic review and a qualitative meta-synthesis? Was it a thematic analysis? This would be important to detail in the paper. Limitations and conclusions are appropriate.

REVIEWER	KLUGAR, Milolav Masaryk University Faculty of Medicine, Czech National Centre for Evidence-Based Healthcare and Knowledge Translation (Cochrane Czech Republic, Czech EBHC: JBI Center of Excellence, Masaryk University GRADE Centre)
REVIEW RETURNED	29-Dec-2020

GENERAL COMMENTS	Dear authors, the topic of your review is of interest and importance, however, I have identified several issues within your manuscript; here are the major issues: 1) Rapid review design is not appropriate for the topic researched. 2) The rationale for conduct the review is missing. 3) A preliminary search of existing reviews and reviews registries is missing. 4) A priory published or registered protocol is missing. 5) Several review questions are formulated, however, which research question informed the search? What is the main objective of the review? 6) Were meta-analyses not possible to be performed?

VERSION 1 – AUTHOR RESPONSE

Reviewer #1: Dr. Julie Langan, University of Glasgow

Comments to the Author: This was a very interesting paper about ward design and how it may impact on rates of seclusion and restraint. It is clinical relevant. They are a number of strengths of paper but there are also a number of limitations that should be addressed prior to publication.

• The methods were not completely clear. What are the inclusion and exclusion criteria for the studies, other than non-English language?

The authors have clarified inclusion and exclusion in the method section (page 6):

'The following academic databases were searched: Cochrane Library, Medline, PsycINFO, Scopus and Avery. Additional literature was identified from the expert knowledge of academics on the research team. Inclusion criteria were as follows: studies that directly reported the impacts of physical design features of mental health inpatient facilities on the use of seclusion and physical restraint; mental health inpatient settings including adult and child and adolescent services, psychiatric intensive care units (PICU's) and forensic mental health inpatient units; studies published between January 2010 to 28 August 2019; English-language only. Non-peer reviewed studies and literature reviews were excluded, but their references were used to identify additional literature.'

• Would it be helpful to include a summary table of the 35 included studies, such as country, ward type (adult, LD etc), study design etc. It looks like (from the results) that adult wards and CAMHS wards were included. But were PICU's, Older Adults Wards, Forensic and LD wards included too? I think more description about the 35 studies would be helpful as they are clearly very varied.

The authors have included an additional column that describes the country in which the study took place in Table 2, edited the initial study type column to represent study. The inclusion criteria on page 6 clarify which mental health inpatient settings were included and the population/setting column in Table 2 was edited to clarify the settings.

We have added the following additional study details in text on page 7:

'Sixteen studies involved a pre-post study, seven studies were qualitative studies and three studies used a mixed-method approach, four were retrospective cohort studies and one study was a prospective cohort study, three studies were case-control studies and one study used a Delphi method. The studies were performed in a variety of settings including 12 inpatient psychiatric facilities, one university clinic, eight acute inpatient settings, six Psychiatric Intensive Care Units (PICUs), three

child and/or adolescent inpatient settings, three forensic inpatient settings and one inpatient setting for older people. Additionally, two studies included consumers with lived experience of restraint and/or seclusion and their supporters (see Table 2).'

• I wonder if there was any comment on ward occupancy as this is also relevant.

The authors acknowledge that ward occupancy may also be relevant in efforts to reduce the use of restraint and seclusion, which was included in our search terms. However, this did not come up as a key theme when collating and synthesising the results across studies.

• It was also unclear how the over-arching themes were identified? Is this a rapid systematic review and a qualitative meta-synthesis? Was it a thematic analysis? This would be important to detail in the paper.

The results from the various studies were categorised and collated using Table 2. Subsequently, this informed our synthesis of the evidence which identified the overarching themes across studies (both qualitative and quantitative). This descriptive approach is typical for rapid reviews and has previously been described by Khangura et al. (2012).

To clarify this, we noted the following on page 7:

'After data from the included studies were extracted, categorised and collated, we synthesised the results and identified overarching themes across studies (21).'

Khangura, S., Konnyu, K., Cushman, R., Grimshaw, J., & Moher, D. (2012). Evidence summaries: the evolution of a rapid review approach. Systematic reviews, 1(1), 1-9.

• Limitations and conclusions are appropriate.

We thank the reviewer for this comment and their time and effort.

Reviewer #2: Dr. Milolav KLUGAR, Masaryk University Faculty of Medicine

Comments to the Author: Dear authors, the topic of your review is of interest and importance, however, I have identified several issues within your manuscript; here are the major issues:

• Rapid review design is not appropriate for the topic researched.

In response to the reviewer's comments we have considered the appropriateness of a rapid review. We found that rapid reviews are recognised as a valid design to research topics that address urgent and emergent health issues and questions deemed to be of high priority (Cochrane Rapid Reviews, Garritty 2020).
Reducing and ultimately eliminating the use of seclusion and restraint is of high priority, as it's use is widely acknowledged as being traumatic and risk-focused. The authors argue that the importance of the physical environment in reducing the use of seclusion and restraint is an emerging health issue and is highly relevant for future evidence-based policy making and practice. This is also evident by the fact that the current research project was tendered by the Ministry of Health (NSW) to be used in co-design and facility and infrastructure planning processes.
A previous study has shown that conclusions do not differ extensively between rapid versus full systematic reviews (Watt et al 2008). Our rapid review involved a specific and defined question and the authors have made an effort to be transparent in terms of methods used and limitations or biases introduced by the use of a more streamlined approach, as discussed in the method and limitation

section. Additionally, we have performed a critical assessment to assess the methodological quality of the studies included, as recommended (Garritty 2020).

We included the following on page 5, clarifying the need for a rapid review as requested by the Ministry of Health:

'The importance of the physical environment in reducing the use of seclusion and restraint is an emerging health issue with relevance for future evidence-based policy making and practice. A rapid review was conducted to summarise current evidence on this topic in a timely manner to inform future co-design, facility and infrastructure planning processes (19). This rapid review aimed to provide an overview of the current research literature of architectural design features of mental health facilities that can help reduce the use of seclusion and restraint.'

'This rapid review was brokered by the Sax Institute for the NSW Ministry of Health, as such the research team received the specific research topic and aim based on the needs of the NSW Ministry of Health. A comprehensive search strategy for academic literature a priori was developed by two researchers (SO, CM), a research librarian (EL) and with advice from a senior researcher (LB), as well as input from the commissioning agency.'

References:
Garritty, C., Gartlehner, G., Nussbaumer-Streit, B., King, V. J., Hamel, C., Kamel, C., ... & Stevens, A. (2020). Cochrane Rapid Reviews Methods Group offers evidence-informed guidance to conduct rapid reviews. Journal of Clinical Epidemiology.

Watt, A., Cameron, A., Sturm, L., Lathlean, T., Babidge, W., Blamey, S., ... & Maddern, G. (2008). Rapid reviews versus full systematic reviews: an inventory of current methods and practice in health technology assessment.

• The rationale for conduct the review is missing.

As the introduction outlines there is emerging evidence that key design features for mental health facilities may impact on broader mental health outcomes and consumer experiences – and an association between the physical characteristics of the environment and a reduction in the use of seclusion and restraint has previously been highlighted. Following on from this, the current review aimed to provide a review of the research literature of architectural design features of mental health facilities that can help reduce the use of seclusion and restraint.

The authors agree with the reviewer that the rationale for the review can be stated more clearly in the introduction. The introduction section has been restructured to clarify the rationale. The changes are highlighted on page 4 – 5.

• A preliminary search of existing reviews and reviews registries is missing.

We identified three reviews in our search that where relevant to our topic, however they did not directly target our research question. As noted on page 6: '[…] literature reviews were excluded, but their references were used to identify additional literature.'

We reference each of these reviews in our introduction. One review identified key design features for mental health facilities that may impact on broader mental health outcomes and consumer experiences (Connellan et al, 2013), one review specifically looked the emergency department environment which was out of scope (Liddicoat, 2019), and another review did not look at design

specifically but identified practices related to the physical environment that could contribute to being trauma-informed (Muskett, 2014).

Previous reviews we identified and referenced include:

Connellan K, Gaardboe M, Riggs D, Due C, Reinschmidt A, Mustillo L. Stressed spaces: mental health and architecture. HERD: Health Environments Research & Design Journal. 2013;6(4):127-68.

Liddicoat S. Designing a supportive emergency department environment for people with self harm and suicidal ideation: A scoping review. Australasian Emergency Care. 2019;22:22.

Muskett C. Trauma-informed care in inpatient mental health settings: A review of the literature. International journal of mental health nursing. 2014;23(1):51-9.

• A priory published or registered protocol is missing.

The authors recognise that a registered research protocol would be preferred. However, the current research project was a short-term project funded by the Ministry of Health (NSW) to be used in co-design and facility and infrastructure planning processes. As such, our timeline did not allow for a priory protocol registration. It is generally recognised that rapid reviews streamline traditional systematic review methods in order to synthesize evidence within a shortened timeframe (Hamel et al, 2020).

Hamel, C., Michaud, A., Thuku, M., Skidmore, B., Stevens, A., Nussbaumer-Streit, B., & Garritty, C. (2020). Defining Rapid Reviews: a systematic scoping review and thematic analysis of definitions and defining characteristics of rapid reviews. Journal of Clinical Epidemiology.

We have added the following in our limitations section:

‘The authors recognise that a priory registration and publication of the study protocol is missing, which was not feasible due to time constraints. It is generally recognised that rapid reviews streamline traditional systematic review methods to synthesize evidence within a shortened timeframe (65).’

• Several review questions are formulated, however, which research question informed the search? What is the main objective of the review?

The introduction section has been restructured to clarify the rationale and aim of the study (highlighted on page 4 -5). Page 5 in the method section also notes: ‘This included studies that directly reported the impacts of physical design features of mental health facilities on the use of seclusion and physical restraint.’

• Were meta-analyses not possible to be performed?

The authors recognise that a meta-analysis would be useful, however point out that the current research project was a short-term project funded by the Ministry of Health (NSW) to be used in co-design and facility and infrastructure planning processes. As such, our timeline did not allow for a meta-analysis to be performed. Furthermore, the authors would point out that a meta-analysis would not have been possible because the included studies were heterogeneous in terms of the populations, methods and outcomes tested.

VERSION 2 – REVIEW

REVIEWER	Langan, Julie University of Glasgow, Institute of Health and Wellbeing
REVIEW RETURNED	10-Jun-2021

GENERAL COMMENTS	I thank the authors for addressing all my comments. These appear to have been done to a satisfactory level. However Figure 1 was missing from the uploaded documents.